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Device and method for preparing bone cement paste

Abstract

A device for preparing a bone cement paste from two starting components, comprising a mixing unit comprising a hollow-cylindrical cartridge with an interior space in which a bone cement powder is stored as the first starting component, a dispensing plunger which is displaceable axially in the interior space and seals the interior space at a proximal cartridge end in terms of fluid conduction, and a mixing rod which is guided into the interior space via a mixing rod duct of the dispensing plunger and is mounted movably in the interior space, and a reservoir for a monomer liquid as the second starting component comprising a conduit that connects the interior space of the mixing unit to the reservoir in terms of fluid conduction, wherein the reservoir comprises a reservoir container with at least one ampoule.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority pursuant to 35 U.S.C. 119(a) to European Application No. 22182103.6, filed Jun. 30, 2022, which application is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(2) The invention relates to a device for preparing a bone cement paste from two starting components, comprising a mixing unit comprising a hollow-cylindrical cartridge with an interior space in which a bone cement powder is stored as the first starting component, a dispensing plunger which is displaceable axially in the interior space and seals the interior space at a proximal cartridge end in terms of fluid conduction, and a mixing rod which is guided into the interior space via a mixing rod duct of the dispensing plunger and is mounted movably in the interior space, and a reservoir for a monomer liquid as the second starting component, comprising a conduit that connects the interior space of the mixing unit to the reservoir in terms of fluid conduction, wherein the reservoir comprises a reservoir container, in which at least one ampoule, closed in terms of fluid conduction, having an ampoule body and an ampoule head, is arranged and the monomer liquid is stored in the ampoule, and a cavity in the region of the ampoule head, wherein the cavity is connected to the conduit in terms of fluid conduction and comprises a connection to the ampoule, wherein the ampoule head is arranged at least in regions in the connection, and the reservoir container comprises a deformable region at least in portions so that tilting of the ampoule about a pivot point against the connection is enabled.

(3) The invention further relates to a method for preparing a bone cement paste using such a device.

BACKGROUND OF THE INVENTION

(4) Considerable efforts are made to identify devices and methods for preparing bone cement paste, by means of which bone cement paste can be prepared simply, reliably, and quickly. One important aspect of preparing bone cement paste is the avoidance of air inclusions, e.g., gas bubbles, in the bone cement. To avoid this, a plurality of vacuum cementing systems have been described, of which the following are mentioned by way of example: U.S. Pat. Nos. 6,033,105 A, 5,624,184 A, 4,671,263 A, 4,973,168 A, 5,100,241 A, WO 99/67015 A1, EP 1020167 A2, U.S. Pat. No. 5,586,821 A, EP 1016452 A2, DE 3640279 A1, WO 94/26403 A1, EP 1005901 A2, EP 1886647 A1, U.S. Pat. No. 5,344,232 A.

(5) One development consists of developing cementing systems in which both starting components are stored in separate regions of the mixing systems and are only mixed with one another in the cementing system immediately before the cementing application. Such closed, so-called full prepacked systems are mentioned in the following documents: EP 0 692 229 A1, DE 10 2009 031 178 B3, U.S. Pat. Nos. 5,997,544 A, 6,709,149 B1, DE 698 12 726 T2, EP 0 796 653 A2, U.S. Pat. No. 5,588,745 A.

(6) In WO 2012/038002 A1, a device is described in which a monomer liquid is guided via a conduit from a container through a cartridge head into an interior space of a cartridge filled with a bone cement powder, so that a bone cement paste can be prepared within the cartridge by mixing.

The conduit and the cartridge are connected to one another via a form closure. After opening the container, for example by opening a valve, the monomer liquid flows via the conduit into the interior. A disadvantage of the disclosed structure of the device is that the type of opening of the container can be restricted due to the comparatively unstable connection of container to cartridge. For example, opening the container might lead to buckling of the conduit due to tilting of the ampoule.

(7) Another disadvantage of the device is the complex structure of the cartridge head since the latter is used both for introducing the monomer liquid and for mixing the bone cement paste. The cartridge head is furthermore connected to the cartridge via a thread, which could weaken the structural integrity of the device.

(8) In EP 2 404 864 A1, a device for opening ampoules, comprising an outer container, in which a closed, in terms of fluid conduction, ampoule with an ampoule body and an ampoule head is arranged, and a cavity are described, wherein the cavity comprises a connection to the ampoule and the ampoule head is arranged at least in regions in the connection. The walls of the outer container comprise at least one deformable region so that tilting of the ampoule against the connection is enabled which opens the ampoule in terms of fluid conduction.

(9) EP 3 093 067 A1 describes a device for preparing a bone cement paste from a monomer liquid and a bone cement powder, which device comprises a device for opening ampoules, filled with the monomer liquid, in accordance with EP 2 404 864 A1. After opening the ampoule in terms of fluid conduction, the monomer liquid is conveyed via a conduit through an opening into an interior space of a cartridge filled with the bone cement powder, where it is mixed by means of a mixing device, which extends through a dispensing plunger axially opposite the opening, with the bone cement powder to form a bone cement paste. A disadvantage of the device is that the described device has a comparatively bulky structure due to its mode of operation. In addition, introducing the monomer liquid on one side of the cartridge and mixing the bone cement paste via a side of the cartridge axially opposite thereto reduces the handiness of the device, which is accompanied by an increase in the complexity of the device and also its handling.

(10) There is a desire on the market to further simplify devices for preparing bone cement paste.

(11) Objects

(12) It is an object of the present invention to at least partially overcome one or more of the disadvantages resulting from the prior art.

(13) The invention is especially based on the goal of providing a device which permits simple and application-safe opening of one or more ampoules, in particular glass ampoules, with a monomer liquid for the simple, rapid, and application-safe preparation of a bone cement paste. In particular, the opening of the ampoule or ampoules should take place with as little effort as possible and with avoidance of additional separate tools. Furthermore, the opening of the ampoule with as few components as possible should be enabled. Furthermore, the monomer liquid should be provided as losslessly as possible for the preparation of the bone cement paste. The device should also have as high a structural integrity as possible with as compact a construction as possible.

(14) A further object of the invention is to provide a method with which bone cement can be prepared from two starting components, by means of which at least some of the objects already described are achieved at least in part.

Preferred Embodiments of the Invention

(15) The features of the independent claims contribute to at least partially fulfilling at least one of the aforementioned objects. The dependent claims provide preferred embodiments which contribute to at least partially fulfilling at least one of the objects.

(16) A first embodiment of the invention is a device for preparing a bone cement paste from two starting components, comprising a mixing unit comprising a hollow-cylindrical cartridge with an interior space in which a bone cement powder is stored as the first starting component, a dispensing plunger which is displaceable axially in the interior space and seals the interior space at a proximal

cartridge end in terms of fluid conduction, and a mixing rod which is guided into the interior space via a mixing rod duct of the dispensing plunger and is mounted movably, in particular axially displaceably in the interior space and rotatably about a longitudinal axis of the mixing rod, in the interior space, and a reservoir for a monomer liquid as the second starting component, comprising a conduit that connects the interior space of the mixing unit to the reservoir in terms of fluid conduction, wherein the reservoir comprises a reservoir container, in which at least one ampoule, closed in terms of fluid conduction, having an ampoule body and an ampoule head, is arranged and the monomer liquid is stored in the ampoule, and a cavity in the region of the ampoule head, wherein the cavity is connected to the conduit in terms of fluid conduction and comprises a connection to the ampoule, wherein the ampoule head is arranged at least in regions in the connection, and the reservoir container comprises a deformable region at least in portions so that tilting of the ampoule about a pivot point against the connection is enabled, characterized in that the conduit extends through a conduit duct of the dispensing plunger into the interior space.

(17) In one embodiment of the device, the conduit and the conduit duct form a first form closure, preferably a reversible first form closure, between the mixing unit and the reservoir. This embodiment is a second embodiment of the invention, which is preferably dependent upon the first embodiment of the invention.

(18) In one embodiment of the device, the reservoir comprises a connecting element in order to connect the reservoir to the mixing unit via a second form closure, preferably via a reversible second form closure. This embodiment is a third embodiment of the invention, which is preferably dependent upon the second embodiment of the invention.

(19) In one embodiment of the device, the connecting element is a clasp. This embodiment is a fourth embodiment of the invention, which is preferably dependent upon the third embodiment of the invention.

(20) In one embodiment of the device, in a side view, the pivot point, the first form closure, and the second form closure form the vertices of a triangle. This embodiment is a fifth embodiment of the invention which is preferably dependent on the third or fourth embodiment of the invention.

(21) In one embodiment of the device, the first form closure, the second form closure, and the pivot point respectively lie on a straight line running parallel to a longitudinal axis of the cartridge, wherein the straight lines have a different straight-line distance from the longitudinal axis of the cartridge. This embodiment is a sixth embodiment of the invention which is preferably dependent on the third to fifth embodiment of the invention.

(22) In one embodiment of the device, the second form closure is formed between the connecting element and the mixing rod. This embodiment is a seventh embodiment of the invention, which is preferably dependent on the third to sixth embodiment of the invention.

(23) In one embodiment of the device, a vacuum connection is arranged on the dispensing plunger, via which vacuum connection the interior space can be connected, in terms of fluid conduction, to a negative-pressure source, such as a vacuum pump. This embodiment is an eighth embodiment of the invention, which is preferably dependent upon one of the preceding embodiments of the invention.

(24) In one embodiment of the device, the conduit is a tube, a hose, or a tube and a hose. This embodiment is a ninth embodiment of the invention, which is preferably dependent on one of the preceding embodiments of the invention.

(25) In one embodiment of the device, the conduit is axially displaceable in the conduit duct. This embodiment is a tenth embodiment of the invention which is preferably dependent on the ninth embodiment of the invention.

(26) In one embodiment of the device, the conduit extends at least over 70%, preferably at least over 80%, more preferably at least over 90%, of an axial interior length of the interior space. This embodiment is an eleventh embodiment of the invention, which is preferably dependent upon the ninth or tenth embodiment of the invention.

(27) In one embodiment of the device, the mixing unit comprises a duct closure, which is preferably present separately, in order to seal the conduit duct in terms of fluid conduction after removal of the conduit. This embodiment is a twelfth embodiment of the invention, which is preferably dependent upon one of the preceding embodiments of the invention.

(28) In one embodiment of the device, the cartridge comprises a cartridge head at a distal cartridge end axially opposite the proximal cartridge end, which cartridge head seals the distal cartridge end in terms of fluid conduction, wherein the cartridge head comprises a cartridge head duct for connecting, in terms of fluid conduction, a dispensing spout, via which the bone cement paste can be dispensed from the interior space after the preparation, and wherein the cartridge and the cartridge head are designed in one piece. This embodiment is a thirteenth embodiment of the invention, which is preferably dependent upon one of the preceding embodiments of the invention.

(29) A fourteenth embodiment of the invention is a method for preparing a bone cement paste from two starting components by means of a device in accordance with one of the preceding embodiments of the invention, comprising the steps of: a. opening, in terms of fluid conduction, the at least one ampoule by tilting the ampoule, stored in the reservoir container, about the pivot point against the connection, b. flowing of the monomer liquid from the at least one ampoule into the cavity, c. conveying the monomer liquid from the cavity via the conduit into the interior space.

(30) In one embodiment of the method by means of a device in accordance with a third to thirteenth embodiments of the invention, the method additionally comprises the steps of: d. detaching the second form closure, e. detaching the first form closure by pulling out the conduit from the conduit duct, f. sealing, in terms of fluid conduction, the conduit duct, g. mixing bone cement powder and monomer liquid.

(31) This embodiment is a fifteenth embodiment of the invention, which is preferably dependent upon the fourteenth embodiment of the invention.

(32) General

(33) In the present description, range specifications also include the values specified as limits. A specification of the type “in the range of X to Y” with respect to a variable A consequently means that A can assume the values X, Y and values between X and Y. Ranges delimited on one side of the type “up to Y” for a variable A accordingly mean, as a value, Y and less than Y.

(34) Some of the described features are linked to the term “substantially.” The term “substantially” is to be understood as meaning that, under real conditions and manufacturing techniques, a mathematically exact interpretation of terms such as “superimposition,” “perpendicular,” “diameter,” or “parallelism” can never be given exactly, but only within certain manufacturing-related error tolerances. For example, “substantially perpendicular axes” form an angle of 85 degrees to 95 degrees relative to one another, and “substantially equal volumes” comprise a deviation of up to 5% by volume. An “apparatus consisting substantially of plastic material” comprises, for example, a plastics content of ≥ 95 to $\leq 100\%$ by weight. “Essentially adding all of a volume B” comprises, for example, adding ≥ 95 to ≤ 100 vol % of the total volume of B. “Essentially conveying all of a component C” comprises, for example, conveying ≥ 90 to ≤ 100 vol %, and in particular ≥ 95 to ≤ 100 vol %, of the total volume of C.

(35) The terms “proximal” and “distal” are used only to designate the spatially opposite ends of the device or other structural units of the device, and do not permit any inference as to orientation with respect to a human body, such as a user of the device. “Distally to . . .” and “proximally to . . .” or similar formulations correspondingly express only the spatial arrangement of two structural units of the device in relation to one another.

DETAILED DESCRIPTION

(36) A first subject matter of the invention relates to a device for preparing a bone cement paste from two starting components, comprising a mixing unit comprising a hollow-cylindrical cartridge with an interior space in which a bone cement powder is stored as the first starting component, a dispensing plunger which is displaceable axially in the interior space and seals the interior space at

a proximal cartridge end in terms of fluid conduction, and a mixing rod which is guided into the interior space via a mixing rod duct of the dispensing plunger and is mounted movably, in particular axially displaceably in the interior space and rotatably about a longitudinal axis of the mixing rod, in the interior space, and a reservoir for a monomer liquid as the second starting component, comprising a conduit that connects the interior space of the mixing unit to the reservoir in terms of fluid conduction, wherein the reservoir comprises a reservoir container, in which at least one ampoule, closed in terms of fluid conduction, having an ampoule body and an ampoule head, is arranged and the monomer liquid is stored in the ampoule, and a cavity in the region of the ampoule head, wherein the cavity is connected to the conduit in terms of fluid conduction and comprises a connection to the ampoule, wherein the ampoule head is arranged at least in regions in the connection, and the reservoir container comprises a deformable region at least in portions so that tilting of the ampoule about a pivot point against the connection is enabled, characterized in that the conduit extends through a conduit duct of the dispensing plunger into the interior space.

(37) The device serves to mix a bone cement paste from a bone cement powder and a monomer liquid, wherein prior to mixing, the bone cement powder is stored in a mixing unit of the device and the monomer liquid is stored in a reservoir of the device.

(38) The reservoir serves to store the monomer liquid before it prepares the bone cement paste by mixing with the bone cement powder in the mixing unit. For this purpose, the reservoir comprises a reservoir container for receiving one or more, preferably two, closed, in terms of fluid conduction, ampoules, in particular glass ampoules, filled with the monomer liquid and comprising an ampoule head and an ampoule body. The reservoir container surrounds the at least one ampoule, in particular at least the ampoule body, so that the ampoule can be stored securely in the reservoir until it is used. The reservoir container can, for example, be present in the form of a hollow cylinder into which the at least one ampoule is inserted, wherein, for improved transport capability of the device, the reservoir container is shaped such, for example by the reservoir container comprising a cover, that the ampoule cannot escape unintentionally from the device, in particular from the reservoir. The reservoir container is preferably shaped such that two ampoules, in particular two ampoules next to one another, can be stored, preferably with substantially parallel longitudinal axes, in the reservoir.

(39) Furthermore, the reservoir serves to open, in terms of fluid conduction, the at least one ampoule. For this purpose, the reservoir comprises a cavity which is connected via a connection to the ampoule arranged in the reservoir container. The ampoule is stored in the reservoir in such a way that the ampoule head points in the direction of the cavity, whereas the ampoule body is arranged at least partially, preferably entirely, in the reservoir container. The connection extends between cavity and reservoir container, and in fact in such a way that the ampoule head is arranged at least in regions in the connection. For this purpose, the connection has a connection diameter which allows the ampoule head to be inserted into the connection at least in portions. In one embodiment, the connection has a connection diameter which allows the ampoule head to be inserted completely into the connection. The connection diameter is preferably smaller than the diameter of the ampoule body so that the latter cannot be inserted into the connection. For example, the connection is designed as a ring or hollow cylinder, and the ampoule head is surrounded at least in regions by this ring or hollow cylinder. The connection has a structural integrity which exceeds a structural integrity of the ampoule, so that the ampoule can break when it is pressed against the connection.

(40) In order to open the ampoule or, given the presence of two or more ampoules, all ampoules, the reservoir container comprises a deformable region at least in portions, in particular adjacent to a transition of the ampoule head to the ampoule body of the ampoule. In one embodiment, the reservoir container is completely deformable. The deformable region allows tilting of the ampoule about a pivot point against the connection. The connection diameter is in this case matched to the ampoule head in such a way that, during tilting, at least the ampoule body end facing away from

the ampoule head is tilted about the pivot point while at least the ampoule head end facing away from the ampoule body remains within the connection, so that the ampoule is opened in terms of fluid conduction by at least partial bursting of the ampoule, in particular in the region of an ampoule neck between ampoule head and ampoule body. The connection in this case serves primarily to fix the ampoule head against a tilting movement of the ampoule about the pivot point. For example, the connection diameter is not more than 10% larger than the diameter of the ampoule head so that a relatively slight tilting of the ampoule already leads to its opening in terms of fluid conduction.

(41) After opening of the at least one ampoule in terms of fluid conduction, the monomer liquid can flow out of the ampoule into the cavity. The cavity is connected in terms of fluid conduction via a conduit to the mixing unit, in particular to an interior of the cartridge of the mixing unit, in which the bone cement powder is stored. Conveying of the monomer liquid from the cavity into the mixing unit via the conduit can, for example, be triggered via the force of gravity; via a negative pressure in the mixing unit, in particular in the interior of the cartridge; or a combination thereof.

(42) The reservoir can consist of a wide range of materials or material combinations. Examples, the reservoir may consist of a polymer. The polymer is preferably a transparent polymer since, in this way, the user can visually monitor a proper functioning of the reservoir, in particular an outflow of the monomer liquid from the at least one ampoule, during use.

(43) The mixing unit serves to mix the bone cement paste from the bone cement powder and the monomer liquid after conveying the monomer liquid into the mixing unit, in particular into the interior of the mixing unit.

(44) The mixing unit comprises a hollow cylindrical cartridge in which the bone cement powder is stored. A hollow cylindrical cartridge is to be understood as a tubular receptacle which comprises an interior space and a cartridge wall surrounding the interior space. The cross-section of the cartridge can assume any shape. Because the device is easy to manufacture and safe to use, the cross-section, and preferably also the cross-section of the interior space, is of circular design. This allows easy handling for the user and reduces a risk of movable parts wedging within the device, due to the absence of edges.

(45) The mixing unit, in particular the cartridge, can consist of a wide range of materials or material combinations. Examples, the mixing unit, in particular the cartridge, can consist of a polymer. The polymer is preferably a transparent polymer since, in this way, the user can visually monitor a proper functioning of the mixing unit during use.

(46) The mixing unit comprises a dispensing plunger, which is displaceable axially in the interior space and is preferably reversibly axially displaceable and which seals the interior space of the cartridge in terms of fluid conduction at a proximal cartridge end so that no bone cement powder can escape unintentionally from the proximal cartridge end. The dispensing plunger is mounted in the interior space such that it can be advanced in the direction of the distal cartridge end axially opposite the proximal cartridge end. This can serve, for example, to dispense the bone cement paste prepared in the device, in particular in the interior space of the mixing unit, from the distal cartridge end.

(47) The dispensing plunger comprises, preferably centered, a mixing rod duct which extends axially through the dispensing plunger and in which a mixing rod is arranged. The mixing rod in this case terminates with the mixing rod duct in such a way that no bone cement powder or bone cement paste can escape from the mixing rod duct.

(48) The mixing rod serves to mix bone cement powder and monomer liquid in order to prepare the bone cement paste. For this purpose, the mixing rod extends movably, in particular axially displaceably in the interior space and preferably also rotatably in the mixing rod duct about its own axis, through the mixing rod duct into the interior space. The mixing rod preferably has a mixing rod length which is longer than an interior length of the interior space so that the mixing rod can mix the starting components over the entire interior length and at the same time allows the mixing

rod to be operated by a user from outside the mixing unit.

(49) At a distal mixing rod end protruding into the interior space, the mixing rod preferably comprises a mixing element in order to be able to thoroughly mix the starting components over the entire cross section of the interior space, if possible. For example, the mixing element is formed as a mixing disk with multiple, for example two to six, mixing disk ducts running axially through the mixing disk. The mixing rod preferably has a diameter that substantially corresponds to a diameter of the cartridge, which facilitates thorough mixing of the starting components.

(50) In order to facilitate operation from outside the mixing unit for a user, it is preferred that the mixing rod comprises a handle at the proximal mixing rod end not mounted in the interior space.

(51) After the bone cement paste has been prepared, the mixing rod is no longer required for further use of the device. In one embodiment, the mixing rod therefore has a predetermined breaking point at which the mixing rod can be broken off in a controlled manner so that it no longer protrudes from the dispensing plunger. Preferably, the distance of the predetermined breaking point from the distal mixing rod end corresponds approximately to an axial extension of the dispensing plunger. As a result, after the mixing rod has been pulled out of the interior space until the mixing element abuts distally on the dispensing plunger, the predetermined breaking point is located in the region of the proximal dispensing-plunger side facing away from the interior space, which substantially allows complete breaking off of the portion of the mixing rod protruding from the mixing unit. At the same time, the mixing element is thereby arranged directly on the dispensing plunger so that advancing the latter in the direction of the distal cartridge end is not hindered by the mixing rod and/or the mixing element.

(52) The mixing unit is connected in terms of fluid conduction to the reservoir, in particular to the cavity of the reservoir, via the conduit, which extends through a conduit duct of the dispensing plunger at the proximal cartridge end. "In terms of fluid conduction" means that liquids, in particular the monomer liquid, and gases can be exchanged between the reservoir and the mixing unit via the conduit. A filter means, in particular a pore disk, for example made of sintered polypropylene particles, of sintered or compressed polyethylene fibers, of cellulose felt, or of paperboard, is preferably arranged within the conduit, which filter means makes the conduit impermeable to solids. This prevents possible fragments of the at least one ampoule from being able to pass into the interior. The filter means is preferably arranged on the end of the conduit opposite the cavity, so that no bone cement powder can penetrate into the conduit, and the conduit can thus seal in terms of fluid conduction upon contact with the monomer liquid to form the bone cement paste. One such filter means can also be arranged in the cavity in order to already capture possible fragments of the at least one ampoule before, in spatial terms, the conduit.

(53) Both the conveying of the monomer liquid and the mixing of bone cement powder and the conveyed monomer liquid are made possible spatially by the dispensing plunger. This allows the simplest and most compact design possible of the device. At the same time, this extensive concentration of the components required for preparing the bone cement paste in the region of the proximal cartridge end allows a device with the highest possible structural integrity since other regions of the device can be oriented in a targeted manner toward high stability.

(54) One embodiment of the device is characterized in that the conduit and the conduit duct form a first, preferably reversible, form closure between the mixing unit and the reservoir. This allows the monomer liquid to be conveyed in an application-safe and controlled manner from the reservoir into the interior space of the mixing unit for a user of the device.

(55) One embodiment of the device is characterized in that the reservoir comprises a connecting element in order to connect the reservoir to the mixing unit, preferably reversibly, via a second form closure.

(56) In this embodiment, the mixing unit and the reservoir are connected to one another via at least two form closures. The first form closure is formed between the conduit duct of the mixing unit and the conduit of the reservoir, and the second form closure is formed between a connecting element

of the reservoir and the mixing unit.

(57) Upon opening the at least one ampoule by tilting about the pivot point against the connection, a sufficiently large force must be exerted on the ampoule in order to overcome the structural integrity of the ampoule. This force is additionally increased given use of more than one ampoule, such as preferably two ampoules, if the latter are to be opened simultaneously, as is preferred, by tilting about the pivot point against the connection. This force acts transferred on the contact points of reservoir and mixing unit.

(58) If the reservoir and the mixing unit were connected only via the first form closure, the force for opening the at least one ampoule would act entirely on the conduit so that buckling or even tearing off of the conduit could occur, whereby a substantially complete conveying of the monomer liquid from the reservoir into the mixing unit, in particular the interior space, would be hindered or even prevented.

(59) The second form closure between the connecting element and the mixing unit ensures a force distribution of the force, required to open the at least one ampoule, onto the two form closures so that the risk of the device being damaged upon opening the at least one ampoule by tilting is reduced.

(60) Via the two form closures, the reservoir is connected stably to the mixing unit in such a way that the at least one ampoule can be opened by tilting about the pivot point against the connection without damaging the device, and without requiring additional aids in addition to the device to open the at least one ampoule.

(61) In one embodiment, the connecting element is integrally connected to the rest of the reservoir, in particular the connection, the cavity, and/or the reservoir container. In a further embodiment, the connecting means can be detachably connected to the rest of the reservoir, in particular to the connection, the cavity, and/or the reservoir container.

(62) The two form closures may be designed such that a reversible, in particular non-destructive, separation of reservoir and mixing unit is not possible.

(63) One embodiment of the device is characterized in that the first form closure and the second form closure are designed to be detachable. This allows a simple, preferably reversible, non-destructive separation of reservoir and mixing unit so that, after conveying the monomer liquid from the at least one cartridge via the conduit into the interior of the cartridge, the reservoir can be simply separated from the mixing unit. Since the reservoir is no longer required after the conveying of the monomer liquid into the mixing unit, it can facilitate preparation of the bone cement paste for a user, for example by improved handling of the mixing unit.

(64) The connecting element can be designed differently in order to form the second form closure between reservoir and mixing unit.

(65) One embodiment of the device is characterized in that the connecting element is a clasp. The clasp is preferably formed from two clasp notches on a reservoir outer surface, for example an outer surface of the reservoir container, of the cavity or of the connection and from a reversibly detachable clasp element, wherein the clasp element comprises two protuberances which can be inserted into the two clasp notches so that the clasp surrounds the mixing unit, preferably the mixing rod of the mixing unit, in a collar-like manner and thus forms the first form closure. This allows a fast and simply designed and detachable second form closure which is stable and allows safe opening of the at least one ampoule by tilting about the pivot point.

(66) The pivot point as well as the first form closure and the second form closure may be arranged spatially in different ways with respect to one another.

(67) One embodiment of the device is characterized in that, in a side view, in particular a side view of the device, the pivot point, the first form closure, and the second form closure form the vertices of a triangle. In this embodiment, the pivot point and the two form closures lie in a common plane but are not arranged on a straight line running in this plane. The longitudinal axis of the cartridge preferably lies within this plane or runs at least parallel to this plane. The arrangement in the form

of a triangle improves the force distribution of the force required for the opening of the at least one ampoule in terms of fluid conduction, in particular given a tilting movement, used for this purpose, of the ampoule about the pivot point within or parallel to the plane of the triangle. Furthermore, such an arrangement fixes and does not displace the pivot point in this plane, which facilitates a reproducible opening of the at least one ampoule.

(68) The pivot point as well as the first form closure and the second form closure may have different distances from one another.

(69) One embodiment of the device is characterized in that the second form closure has a shorter distance from the first form closure than the pivot point. In this embodiment, the distance between pivot point and first form closure is thus greater than the distance between second form closure and first form closure. In particular given an arrangement of pivot point and the two form closures in the form of a triangle, this allows both improved force distribution of the force, required during tilting to open the ampoule, onto the two form closures and simultaneously an optimally space-saving design of the device. The latter facilitates in particular the handling of the device by a user.

(70) The pivot point and the two form closures in this case preferably form the vertices of a triangle in a side view, wherein the pivot point and the second form closure have the smallest value of the three possible distances between the mentioned points. This leads to a further improvement of the force distribution onto the two form closures and allows a further space-saving design of the device.

(71) One embodiment of the device is characterized in that the first form closure, the second form closure, and the pivot point respectively lie on a straight line running parallel to a longitudinal axis of the cartridge, wherein the straight lines have a different straight-line distance from the longitudinal axis of the cartridge. This arrangement improves the force distribution of the force, required during tilting to open the ampoule, onto the two form closures and also simultaneously allows an optimally space-saving design of the device. The latter facilitates in particular the handling of the device by a user.

(72) The pivot point and the two form closures in this case preferably form the vertices of a triangle, wherein the triangle lies in a plane in which the longitudinal axis of the cartridge also lies, or to which the longitudinal axis of the cartridge runs at least in parallel.

(73) The second form closure between the reservoir, in particular between the connecting element of the reservoir, and the mixing unit can take place at different points of the mixing unit.

(74) One embodiment of the device is characterized in that the second form closure is formed between the connecting element and the mixing rod. This facilitates an attachment of the reservoir to the side of the dispensing plunger opposite the interior space and thus facilitates a conveying of the monomer liquid through the conduit into the interior space. The latter is in particular facilitated if utilizing the force of gravity for conveying the monomer liquid. In addition, such an arrangement of the second form closure allows an optimally simple and space-saving design of the device.

(75) In order to prevent an unwanted advancement of the dispensing plunger, a catch means can be arranged at the dispensing plunger so that the dispensing plunger engages with the cartridge, in particular with the cartridge wall, until the catch means is actively detached by a user of the device.

(76) One embodiment of the device is characterized in that a vacuum connection is arranged on the dispensing plunger, via which vacuum connection the interior space can be connected in terms of fluid conduction to a negative-pressure source, such as a vacuum pump. The vacuum connection thus allows application of a negative pressure in the interior, which negative pressure may, for example, be used to convey the monomer liquid from the reservoir, in particular from the cavity of the reservoir, through the conduit into the interior. This allows simple and rapid conveying of the monomer liquid into the interior. Furthermore, the negative pressure in the interior can be applied or maintained during the mixing of the bone cement paste from the two starting components. Air inclusions in the bone cement paste, which could have a disadvantageous effect on the cured bone cement, can thereby be reduced or avoided.

(77) The conduit can be designed differently in order to conduct the monomer liquid into the interior of the cartridge of the mixing unit.

(78) One embodiment of the device is characterized in that the conduit is a tube, a hose, or a combination of a tube and a hose. This allows a simple production as well as easy and space-saving arrangement of the conduit within the interior. The conduit is preferably a tube.

(79) It is preferred that the conduit, in particular in the form of a tube and/or a hose, has a fluid-conducting internal conduit diameter in a range of 0.5 mm to 2 mm, preferably in a range of 0.5 mm to 1.5 mm. Smaller diameters slow down the conveyance of the monomer liquid. Larger diameters make it more difficult to convey basically all of the monomer liquid, due to lower capillary effects in the conduit.

(80) One embodiment of the device is characterized in that the conduit is axially displaceable in the conduit duct, preferably in the interior space of the cartridge. This allows simple detachment of the first form closure by pulling the conduit out of the conduit duct. The conduit duct preferably surrounds the conduit in a collar-like manner, wherein the conduit is preferably a tube. This represents a stable first form closure which can be detached in a controlled and simple manner by pulling the conduit out of the conduit duct. In order to facilitate detachment of the first form closure by pulling out the conduit from the conduit duct, it is more preferred that the conduit is a flexible tube.

(81) The conduit extends in or through the conduit duct and can extend, in particular axially, to different extents into the interior space of the cartridge. In one embodiment, the conduit extends through the conduit duct and terminates flush therewith on the side of the interior space. In this embodiment, the conduit does not extend within the interior.

(82) One embodiment of the device is characterized in that the conduit extends at least over 70%, preferably at least over 80%, more preferably at least over 90%, of the axial interior length of the interior space. Given spatial orientation of the device with the proximal cartridge end upward and correspondingly the distal cartridge end downward, so that bone cement powder is stored in the vicinity of the distal cartridge end, this allows conveying of the monomer liquid directly into the bone cement powder which is stored in the interior and preferably does not fill the entire interior of the cartridge but rather, for example, only 20-70% by volume of the entire volume of the interior. In this respect, it is preferred that the amount of bone cement powder is selected such that, in the vertical orientation of the cartridge with the cartridge head upward, the conduit can guide the monomer liquid at least onto a surface of the bone cement powder, preferably at least 1 cm deep into the bone cement powder. This facilitates mixing of the starting components of the bone cement paste. Furthermore, the monomer liquid is thereby released in the spatial vicinity of the distal cartridge end in the interior space and thus at a comparatively large spatial distance from the vacuum connection on the dispensing plunger, if present. It is thereby prevented that when negative pressure is applied to the vacuum connection, the monomer liquid is sucked into the vacuum connection and is thus not available for mixing with the bone cement powder.

(83) One embodiment of the device is characterized in that the mixing unit comprises a duct closure in order to seal the conduit duct in terms of fluid conduction after removal of the conduit, preferably by pulling the conduit out of the conduit duct. The duct closure can be present as a separate component or be designed as part of the dispensing plunger or the cartridge. In one embodiment, the duct closure is designed as a plug which can be inserted into the conduit duct, in particular can be inserted from the proximal dispensing-plunger side. In a further embodiment, the duct closure is formed as a slider which seals the conduit duct in terms of fluid conduction by insertion.

(84) Through the duct closure, mixing of the bone cement from the starting components within the mixing unit is enabled, in particular in the interior space of the cartridge, with applied negative pressure at the vacuum connection. Air inclusions in the bone cement paste, which could have a disadvantageous effect on the cured bone cement, can thereby be reduced or avoided.

(85) The distal cartridge end can be designed differently in order to enable the bone cement paste to be prepared in the interior space. For example, the distal cartridge end can be sealed in terms of fluid conduction with a cover until the bone cement paste has been prepared in the interior space. For this purpose, the cover has a diameter which substantially corresponds to a diameter of the interior space so that the cover can completely seal the interior space in terms of fluid conduction at the distal cartridge end. The cover can, for example, be connectable to the cartridge, in particular reversibly, via a threaded connection or via a bayonet connection. After preparing the bone cement paste, the cover can preferably be removed in order to dispense the bone cement paste from the interior space.

(86) One embodiment of the device is characterized in that the cartridge comprises a cartridge head at a distal cartridge end, which cartridge head seals the distal cartridge end in terms of fluid conduction, wherein the cartridge head comprises a cartridge head duct for connecting, in terms of fluid conduction, a dispensing spout, via which the bone cement paste can be dispensed from the interior space, and wherein the cartridge and the cartridge head are designed in one piece. In order to provide a device with the highest possible structural integrity, the cartridge and the cartridge head are designed in one piece. This saves fastening of the cartridge head, for example via a threaded connection, to the cartridge and thereby reduces weakening of the structural integrity of the device.

(87) The cartridge head comprises a cartridge head duct which allows, in terms of fluid conduction, a connection of the mixing unit to a dispensing spout, a tubular, elongate component for controlled dispensing of the bone cement from the interior space. The cartridge head duct preferably comprises an internal thread which can be screwed together with an external thread of a dispensing spout to form a threaded connection in order to connect the dispensing spout to the interior space in terms of fluid conduction. The cartridge head duct has a diameter which is smaller than the diameter of the interior space. For example, the cartridge head duct has a diameter which corresponds to 5-25% of the diameter of the interior space.

(88) The cartridge head duct is preferably sealed in terms of fluid conduction prior to the attachment of a dispensing spout, preferably via a cartridge head closure with an external thread which interacts in the manner of a form closure and/or force closure with an internal thread of the cartridge head duct and thus reversibly fixes the cartridge head closure in the cartridge head duct.

(89) A further subject matter of the invention relates to a method for preparing a bone cement paste from two starting components by means of a device in accordance with one of the preceding embodiments of the invention, comprising the steps of: a. opening, in terms of fluid conduction, the at least one ampoule by tilting the ampoule, stored in the reservoir container, about the pivot point against the connection, b. flowing of the monomer liquid from the at least one ampoule into the cavity, c. conveying the monomer liquid from the cavity via the conduit into the interior space.

(90) In a step a., the at least one ampoule, wherein one or two ampoules are preferred, is opened in terms of fluid conduction by tilting the ampoule, stored in the reservoir container, about the pivot point against the connection. The tilting is enabled by the region of the reservoir container that is deformable at least in portions. The at least one ampoule is stored in the reservoir in such a way that the ampoule head is arranged at least in regions in the connection of the reservoir so that, during tilting of the ampoule, in particular of the ampoule body in the reservoir container, about the pivot point, the ampoule head is pressed against the connection and its structural integrity is thus overcome so that the monomer liquid can flow out of the at least one ampoule. Upon opening, the ampoule preferably breaks in the region of the ampoule neck between ampoule head and ampoule body.

(91) In a step b., the monomer liquid flows from the ampoule opened in step a. into the cavity of the reservoir. In order to facilitate the flow out of the ampoule, the device is preferably held such that the at least one ampoule assumes an angle in a range of 15-35° to the surface of the earth. Depending on the embodiment of the diameter of the connection, the ampoule head can fall into the

cavity of the reservoir upon opening the ampoule in step a. The cavity is preferably dimensioned such that the ampoule head can be completely accommodated in the cavity and the ampoule head is additionally rotatable in the cavity. Monomer liquid possibly present in the ampoule head can thus flow out into the cavity due to the force of gravity and is thus available for preparing the bone cement paste.

(92) In a step c., the monomer liquid is conveyed from the cavity into the interior via the conduit. In one embodiment of the method, the force of gravity is utilized to convey the monomer liquid into the interior. In a further, preferred embodiment of the method, the conveying is triggered by a negative pressure in the interior space, wherein it is preferred that the negative pressure in the interior space is generated by applying a negative pressure, for example by connection, in terms of fluid conduction, the vacuum connection on the dispensing plunger to a negative-pressure source, such as a vacuum pump. In this embodiment, the monomer liquid is preferably introduced directly into the bone cement powder by means of the conduit so that a suction of the monomer liquid from the interior through vacuum connection and into the negative-pressure source is avoided.

(93) One embodiment of the method is characterized in that the method further comprises the following steps: d. detaching the second form closure, e. detaching the first form closure by pulling out the conduit from the conduit duct, f. sealing, in terms of fluid conduction, the conduit duct, g. mixing bone cement powder and monomer liquid.

(94) This embodiment of the method is carried out by means of one of the above-described embodiments of the device, which comprises the connecting element for connecting the reservoir to the mixing unit via a second form closure.

(95) After the monomer liquid has been conveyed into the mixing unit, there is no longer need to keep the reservoir connected to the mixing unit via the two form closures.

(96) In a step d., the second form closure between the reservoir, preferably the connecting element of the reservoir, and the mixing unit, preferably between the connecting element and the mixing rod of the mixing unit, is detached.

(97) In a step e., the first form closure is detached by pulling the conduit out of the conduit duct.

(98) Steps d. and e. may be carried out in any order, wherein it is preferred to first perform step d. and then step e. since pulling out the conduit in step e. is facilitated when the second form closure has already been detached.

(99) In a step f., the conduit duct is sealed in terms of fluid conduction after the conduit has been pulled out, so that the bone cement paste can be mixed given applied negative pressure. The sealing of the cartridge head duct in terms of fluid conduction preferably takes place by using the duct closure.

(100) In a step g., mixing of the bone cement powder and the monomer liquid takes place to prepare the bone cement paste. The mixing preferably takes place with applied negative pressure in the interior. More preferably, the mixing takes place while the mixing rod is moved up and down.

(101) The prepared bone cement paste can be discharged from the device in different ways. For example, the bone cement paste may be retrieved from the interior with a spatula.

(102) One embodiment of the method is characterized in that, after the bone cement paste has been mixed, a dispensing spout is attached in terms of fluid conduction in a cartridge head duct to the cartridge head of the mixing unit, and the bone cement paste is dispensed through the dispensing spout by advancing the dispensing plunger in the direction of the cartridge head. This allows targeted and controlled dispensing of the bone cement paste from the device.

(103) In order to dispense the bone cement paste from the device, the device is preferably connected to a dispensing aid, in particular a dispensing gun, which displaces the dispensing plunger in the direction of the cartridge head and thus dispenses the bone cement paste from the interior space.

(104) The device is characterized in that it prepares a bone cement paste from two starting components. Bone cement paste is understood to mean a substance that is suitable in the field of

medical technology for creating a stable connection between artificial joints, such as hip and knee joints, and bone material. By curing, a bone cement paste becomes a bone cement. These bone cements are preferably polymethyl methacrylate bone cements (PMMA bone cements). PMMA bone cements have been used for a long time in medical applications and are based upon the work of Sir Charnley (cf. Charnley, J., Anchorage of the femoral head prosthesis of the shaft of the femur. *J. Bone Joint Surg.* 1960; 42, 28-30.). PMMA bone cements can thereby be produced from a bone cement powder as a first starting component and a monomer liquid as a second starting component. With a suitable composition, the two starting components can be storage-stable, separately from one another. When the two starting components are brought into contact with one another, a plastically-deformable bone cement paste is produced by the swelling of the polymer components of the bone cement powder. In this case, polymerization of the monomer by radicals is initiated. As the polymerization of the monomer progresses, the viscosity of the bone cement paste increases until it cures completely.

(105) Bone cement powder is understood to mean a powder that comprises at least one particulate polymethyl methacrylate and/or a particulate polymethyl methacrylate copolymer. Examples of copolymers are styrene and/or methyl acrylate. In one embodiment, the bone cement powder can additionally comprise a hydrophilic additive which supports the distribution of the monomer liquid within the bone cement powder. In a further embodiment, the bone cement powder can additionally comprise an initiator which initiates the polymerization. In a further embodiment, the bone cement powder can additionally comprise a radiopaque material. In yet another embodiment, the bone cement powder can additionally comprise pharmaceutically-active substances, such as antibiotics.

(106) The bone cement powder preferably comprises, as a hydrophilic additive, at least one particulate polymethyl methacrylate and/or a particulate polymethyl methacrylate copolymer, an initiator, and a radiopaque material, or consists of these components. More preferably, the bone cement powder comprises at least one particulate polymethyl methacrylate and/or a particulate polymethyl methacrylate copolymer, an initiator, a radiopaque material, and a hydrophilic additive, or consists of these components. Most preferably, the bone cement powder comprises at least one particulate polymethyl methacrylate and/or a particulate polymethyl methacrylate copolymer, an initiator, a radiopaque material, a hydrophilic additive, and an antibiotic, or consists of these components.

(107) According to the invention, the particle size of the particulate polymethyl methacrylate and/or the particulate polymethyl methacrylate copolymer of the bone cement powder of the sieve fraction can correspond to less than 150 μm , preferably less than 100 μm .

(108) According to the invention, the hydrophilic additive can be designed in particulate and/or fibrous form. In a further embodiment, the hydrophilic additive can be slightly soluble, and preferably insoluble, in methyl methacrylate. In a further embodiment, the hydrophilic additive can have an absorption capacity of at least 0.6 g methyl methacrylate per gram of hydrophilic additive. In a further embodiment, the hydrophilic additive can comprise a chemical substance comprising at least one OH group. In this case, the hydrophilic additive can preferably have covalently-bonded OH groups at its surface. Examples of such preferred hydrophilic additives can be additives selected from the group comprising cellulose, oxycellulose, starch, titanium dioxide, and silicon dioxide, wherein pyrogenic silicon dioxide is particularly preferred. In one embodiment, the particle size of the hydrophilic additive can correspond to the sieve fraction of less than 100 μm , preferably less than 50 μm , and most preferably less than 10 μm . The hydrophilic additive can be contained in an amount of 0.1 to 2.5% by weight, based on the total weight of the bone cement powder.

(109) According to the invention, the initiator can contain dibenzoyl peroxide or consist of dibenzoyl peroxide.

(110) According to the invention, a radiopaque material is understood to mean a substance that makes it possible to make the bone cement visible on diagnostic X-ray images. Examples of

radiopaque materials can include barium sulfate, zirconium dioxide, and calcium carbonate.

(111) According to the invention, the pharmaceutically-active substance can comprise one or more antibiotics and, optionally, added cofactors for the one or more antibiotics. Preferably, the pharmaceutically-active substance consists of one or more antibiotics and, optionally, added cofactors for the one or more antibiotics. Examples of antibiotics include, inter alia, gentamicin, clindamycin, and vancomycin.

(112) According to the invention, the monomer liquid can comprise the monomer methyl methacrylate or consist of methyl methacrylate. In one embodiment, the monomer liquid comprises, in addition to the monomer, an activator dissolved therein, such as N,N-dimethyl-p-toluidine, or consists of methyl methacrylate and N,N-dimethyl-p-toluidine.

(113) The features disclosed for the device are also disclosed for the method, and vice versa.

Description

FIGURES

(1) In the following, the invention is illustrated further, by way of example, by figures. The invention is not limited to the figures.

(2) In the figures:

(3) FIG. 1 shows a schematic longitudinal section of a device for preparing a bone cement paste, comprising a mixing unit and a reservoir with an ampoule filled with a monomer liquid,

(4) FIG. 2 shows a perspective side view of a portion of the device of FIG. 1,

(5) FIG. 3 shows the device of FIGS. 1 and 2, wherein a first form closure, a second form closure, and a pivot point are indicated,

(6) FIG. 4 shows the device of FIGS. 1 to 3 during an opening, in terms of fluid conduction, of the ampoule and during conveying of the monomer liquid into the mixing unit,

(7) FIG. 5 shows the device of FIGS. 1 to 4, wherein the reservoir is separated from the mixing unit and the bone cement paste is prepared in the mixing unit,

(8) FIG. 6 shows the device of FIGS. 1 to 5 during dispensing of the bone cement paste from the mixing unit, and

(9) FIG. 7 shows a flow chart of a method for preparing a bone cement paste.

DESCRIPTION OF THE FIGURES

(10) FIG. 1 shows a schematic longitudinal section of a side view of an exemplary embodiment of a device **100** for preparing a bone cement paste from two starting components, in an initial state. The device **100** comprises a mixing unit **200** and a reservoir **300**, which are connected to one another via two form closures (see also FIG. 3).

(11) The mixing unit **200** is constructed like a tube and comprises a hollow cylindrical cartridge **210** with an interior **215**, in which a bone cement powder **500** is stored as a first starting component. A proximal cartridge end **211** of the cartridge **210** is sealed with a dispensing plunger **280** that can be displaced axially in the interior space **215**, and a distal cartridge end **212** axially opposite the proximal cartridge end **211** is sealed with a cartridge head **220**. In the shown embodiment, the cartridge **210** and the cartridge head **220** are designed in one piece. The dispensing plunger **280** reversibly engages at the proximal cartridge end **211** in order to prevent an unintentional advance thereof in the direction of the distal cartridge end **212** in the course of the preparation of the bone cement paste.

(12) The cartridge head **220** comprises a cartridge head duct **225** which is reversibly sealed with a cartridge head closure **226**. For this purpose, the cartridge head **220** comprises an internal thread in the region of the cartridge head duct **225**, which internal thread engages in a form closure and/or force closure with an external thread of the cartridge head closure **226**.

(13) The dispensing plunger **280** comprises a mixing rod duct **235** which runs axially through the

dispensing plunger **280** and in which a mixing rod **230** is arranged axially displaceably and rotatably about itself. The mixing rod **230** seals the mixing rod duct **235** so that no bone cement powder **500** can unintentionally escape from the cartridge **210** through the mixing rod duct **235**.

(14) At an end of the mixing rod **230** facing away from the interior space **215**, the former comprises a handle **250**. The handle **250** makes it easier for a user to handle the device **100** by facilitating an axial movement of the mixing rod **230** within the interior space **215**. At an end axially opposite the handle **250**, the mixing rod **230** is equipped with a mixing element **245** in the form of a mixing disk which facilitates mixing of the bone cement paste in the interior space **215** via an axial displacement of the mixing rod **230** within the interior space **215**.

(15) A vacuum connection **260** is attached to the dispensing plunger **280**, via which vacuum connection the interior space **215** can be connected in terms of fluid conduction to a negative-pressure source (not shown).

(16) The reservoir **300** comprises a tubular reservoir container **320**, in which an ampoule **330**, in particular a glass ampoule, is stored. In further embodiments, not shown, more than one ampoule **330**, preferably two ampoules **330** next to one another, can be stored in the reservoir **300**. The ampoule **330** comprises an ampoule body **331**, an ampoule head **332** facing the mixing unit **200**, and an ampoule neck **333**, which lies between ampoule body **331** and ampoule head **332** and acts as a predetermined breaking point for the ampoule **330**. A monomer liquid **510** is stored in the ampoule **330** as a second starting component of the bone cement paste. The ampoule head **332** of the ampoule **330** is arranged in portions in a connection **350** which connects a cavity **340** of the reservoir **300** to the ampoule **330**. The connection **350** comprises a connection diameter **355** which is approximately 5% larger than a diameter of the ampoule head **332** so that the connection **350** fixes the ampoule head **332** against tilting within the drawing plane. The reservoir container **320** comprises a deformable region **325** in the region of a transition from connection **350** to ampoule body **331** in order to enable tilting of the ampoule **330**, in particular of the ampoule head **332**, against the connection **350**, wherein, in the shown embodiment, tilting is possible in the drawing plane.

(17) Within the cavity **340**, a filter element **345** is arranged in the reservoir **300** so that after the ampoule **330** has been opened in terms of fluid connection, fragments thereof cannot pass via the cavity **340** into the mixing unit **200** but rather are retained on the filter element **345**.

(18) The cavity **340** is connected in terms of fluid conduction via a conduit **310**, in the form of a tube, to the interior **215** of the cartridge **210**. For this purpose, the conduit **310** extends into the interior space **215** through a conduit duct **290** of the dispensing plunger **280**. The conduit extends approximately over 95% of an interior length **216** of the interior space **215** so that, given the shown orientation of the device **100**, the monomer liquid **510** can be conveyed directly into the bone cement powder **500** via the conduit **310**, and has a sufficiently large distance from the vacuum connection **260** of the mixing unit **200** so that the risk of the monomer liquid **510** being suctioned directly out of the conduit **310** through the vacuum connection **260** from the interior space **215** is reduced.

(19) The conduit duct **290** and the conduit **310** form a first form closure (cf. also FIG. 3) by which the mixing unit **200** and the reservoir **300** are connected to one another.

(20) The reservoir **300** furthermore comprises a connecting element **360** in the form of a clasp (cf. also FIG. 2) which is arranged in a collar-like manner around the mixing rod **230** and forms a second form closure (cf. also FIG. 3) between mixing unit **200** and reservoir **300**.

(21) FIG. 2 shows a schematic side view of a portion of the device **100** of FIG. 1. It is visible in FIG. 2 that the connecting element **360** is formed from two clasp notches **361** on an outer surface of the reservoir **300** and a clasp element **362** that can be detached from the two clasp notches **361**. The clasp element comprises two protuberances (not shown) which can be inserted into the two clasp notches **361** so that the connecting element **360** can be reversibly closed in a collar-like manner around the mixing rod **230** in order to form the second form closure stably and securely

between the mixing unit **200** and the reservoir **300**. In the shown embodiment, the two clasp notches **361** and the two protuberances of the clasp element **362** are of triangular design, wherein an outer diameter of the protuberances substantially corresponds to a respective internal diameter of the clasp notches **261**.

(22) In the view shown, a hose **450** is arranged on the vacuum connection **260** (not visible) and makes the device **100** connectable in terms of fluid conduction to a negative-pressure source (not shown), such as a vacuum pump.

(23) FIG. 3 shows the device **100** of FIGS. 1 and 2, wherein the first form closure **400**, the second form closure **410**, and a pivot point **420** about which, due to the deformable region **325**, the ampoules **330** can be pressed by tilting against the connection **350** (cf. FIG. 1) are indicated by filled circles.

(24) In the shown side view of the device **100**, the first form closure **400**, the second form closure **410**, and the pivot point **420** form a triangle (indicated by connecting lines between the filled circles). When the ampoules **330** are tilted about the pivot point **420** so that the ampoules **330**, in particular the ampoule heads **332** (cf. FIG. 1), are pressed against the connection **350** (cf. FIG. 1), the force required in this case for the opening of the ampoules **330** in terms of fluid conduction is distributed to the first form closure **400** and the second form closure **410**. An advantageous force distribution is achieved via the arrangement of the first form closure **400**, the second form closure **410**, and the pivot point **420** in the form of a triangle. In particular, the risk of the conduit **310** buckling or breaking off can thereby be reduced.

(25) The first form closure **400**, the second form closure **410**, and the pivot point **420** are arranged relative to one another in such a way that the second form closure **410** has a shorter distance from the first form closure **400** than the pivot point **420**. The pivot point **420** and the first form closure **400** are thus spaced farther apart from one another than the first form closure **400** and the second form closure **410**. This ensures improved force distribution of the force, required during tilting about the pivot point **420** to open the ampoules **330**, onto the two form closures **400**, **410** and also, at the same time, ensures an optimally space-saving design of the device **100**. The latter in particular facilitates the handling of the device **100** by a user.

(26) The first form closure **400** lies on a straight line **401** running parallel to a longitudinal axis **213** of the cartridge **210**, the second form closure **410** lies on a further straight line **411** running parallel to the longitudinal axis **213** of the cartridge **210**, wherein the longitudinal axis **213** and the straight line **411** are substantially congruent through the second form closure **410**, and the pivot point **420** lies on a further straight line **421** lying parallel to the longitudinal axis **213** of the cartridge **210**, wherein the straight lines **401**, **411**, **421** all have a different straight-line distance from the longitudinal axis **213** of the cartridge **210**. In the shown embodiment, the straight-line distance between the straight line **421** through the pivot point **420** and the longitudinal axis **213** is greatest, followed by the straight-line distance between the straight line **401** through the first form closure **400**, and the longitudinal axis **213**. The different straight-line distances improve the force distribution of the force, required during tilting about the pivot point **420** to open the ampoules **330**, onto the two form closures **400**, **410**, and at the same time allow an optimally space-saving design of the device **100**. The latter in particular facilitates the handling of the device **100** by a user.

(27) FIG. 4 shows the device **100** of FIGS. 1 to 3 with ampoule **330** tilted about the pivot point **420** (cf. FIG. 3). For this purpose, the reservoir container **320** was bent at the deformable region **325** so that the ampoule head **332** was pressed against the connection **350** and the ampoule **330** was opened in terms of fluid conduction in the region of the ampoule neck (cf. FIG. 1). The monomer liquid **510** stored in the ampoule **330** opened in terms of fluid conduction has already largely flowed out of the ampoule **330** into the cavity **340**. The ampoule head **332** is completely transferred from the connection **350** into the cavity **340**. In this case, the ampoule head **332** was captured by the filter element **345** so that it or fragments thereof cannot pass to or through the conduit **310**. The cavity **340** is dimensioned such that the ampoule head **332** can be mounted so as to be completely

rotatable therein, so that monomer liquid **510** that is possibly still present in the ampoule head **332** after the opening of the ampoule **330** can flow out into the cavity **340**. This is already done in FIG. **4** with ampoule head **332**. In the shown embodiment of the device **100**, the conduit **310**, in particular a diameter of the conduit **310**, is designed such that, due to its surface tension, the monomer liquid **510** cannot flow through the conduit **310** into the interior space **215** of the cartridge **210** solely due to the force of gravity. The monomer liquid **510** that flowed out of the ampoules **330** therefore first collects in the cavity **340** until a user of the device **100** actively intervenes. In order to convey the monomer liquid **510** from the cavity **340** into the interior space **215**, the device **100**, in particular the vacuum connection **260**, is connected in terms of fluid conduction via a hose **450** to a negative-pressure source **460** in the form of a vacuum pump. Via the hose **450** and the vacuum connection **260**, the negative-pressure source **460** generates a negative pressure in the interior space **215**, which negative pressure sucks the monomer liquid **510** from the cavity **340** via the conduit **310** into the mixing unit **200**. Due to the axial extension of the conduit **310**, the monomer liquid **510** is introduced directly into the bone cement powder **500**, which reduces the risk of said monomer liquid being sucked through the vacuum connection **260** into the negative-pressure source **460**. The negative pressure in the interior **215** is maintained at least until the monomer liquid **510** is substantially completely transferred into the mixing unit **200**.

(28) FIG. **5** shows the device **100** from FIGS. **1** to **4** when the bone cement paste **520** is mixed from the two starting components in the interior space **215** of the mixing unit **200**, wherein the reservoir **300** has been removed prior to mixing for better handling. In order to remove the reservoir **300**, the first form closure **400** and the second form closure **410** between mixing unit **200** and reservoir **300** were detached (cf. FIGS. **1** to **4**). In order to detach the second form closure **410**, the clasp element **362** has been pulled out of the two clasp notches **261** (cf. FIG. **2**), and in order to detach the first form closure **400**, the conduit **310** has been pulled out of the conduit duct **290** (cf. FIGS. **1** to **4**). Due to simpler handling, first the second form closure **410** and then the first form closure **400** (cf. FIG. **3**) were detached. After removal of the reservoir **300**, the conduit duct **290** was sealed in terms of fluid conduction by inserting a separate duct closure **270** in the form of a plug, so that the mixing of the bone cement paste **520** can take place with negative pressure applied in the interior space **215**.

(29) In order to mix the bone cement paste **520**, the mixing rod **230** together with the mixing element **240** was repeatedly axially moved up and down in the interior space **215**.

(30) FIG. **6** shows the device **100** of FIGS. **1** to **5** during dispensing of the bone cement paste **520**. To this end, the dispensing plunger **280** which was previously engaged at the proximal cartridge end **211** was detached and advanced by means of a dispensing aid **550** in the form of a dispense gun (only shown in portions) in the direction of the cartridge head **220** at the distal cartridge end **212**. In order to facilitate the advance of the dispensing plunger **280**, in particular when using a dispensing aid **550**, the mixing rod **230** was pulled out of the interior space **215** in advance to such an extent that the mixing element **240** abuts on the dispensing plunger **280** and the part of the mixing rod **230** protruding from the dispensing plunger **230**, together with the handle **250** subsequently breaks off at a predetermined breaking point in the region of the dispensing plunger **280**.

(31) Furthermore, the cartridge head closure **226** was removed and replaced with a dispensing spout **530**. The dispensing spout **530** comprises an external thread which interacts in the manner of a form closure and/or force closure with the internal thread of the cartridge head in order to connect the dispensing spout **530** to the interior space **215** in terms of fluid conduction. The dispensing spout **530** facilitates controlled and accurate dispensing of the bone cement paste **530** from the device **100**.

(32) FIG. **7** shows a flow chart of a method **600** for preparing a bone cement paste **520** from two starting components by means of a device **100** in accordance with FIGS. **1** to **6**, comprising steps **610** to **630**, preferably **610** to **670** and optionally also step **680**.

(33) In a step **610**, the at least one ampoule **330** in the reservoir **300** is opened in terms of fluid conduction by tilting about the pivot point **420** against the connection **350**. The force required for this purpose is preferably distributed onto the two form closures **400**, **410** so that buckling of the conduit **310** is prevented.

(34) In a step **620**, the monomer liquid **510** flows from the at least one ampoule **330**, opened in terms of fluid conduction in step **610**, into the cavity **340** of the reservoir **300**. The outflow of the monomer liquid **510** is preferably caused by the force of gravity. During the outflow of the monomer liquid **510** the ampoule head **332**, further or other fragments of the at least one ampoule **332** are preferably retained by a filter element **345** in the cavity **340** so that it or they cannot penetrate into the conduit **310**.

(35) In a step **630**, the monomer liquid **510** is conveyed from the cavity **340** via the conduit **310** into the interior **215** of the cartridge **210** and to the bone cement powder **500**. In one embodiment, the force of gravity is utilized to convey the monomer liquid **510**. In a further embodiment, a negative pressure applied in the interior **215** is used to convey the monomer liquid **510**, which negative pressure is preferably provided via a negative-pressure source **460**, for example a vacuum pump, connected to the vacuum connection **260**. The conveying of the monomer liquid **510** preferably takes place directly into the bone cement powder **500**, which can bring about a better mixing capability and, at the same time, reduces or prevents a risk of the monomer liquid **510** being sucked from the interior space **215** via the vacuum connection **260** when negative pressure is applied.

(36) In a step **640**, detachment of the second form closure **410** preferably takes place. The connecting element **360** is more preferably a clasp so that the clasp element **362** is pulled out of the two clasp notches **361** in order to detach the second form closure **410**.

(37) In a step **650**, detachment of the first form closure **400** preferably takes place by pulling the conduit **310** out of the cartridge head duct **225**.

(38) By detaching **640**, **650** the two form closures **400**, **410**, the reservoir **300** is separated from the mixing unit **200** and can be removed. This facilitates the handling of the mixing unit **200**.

(39) The two form closures **400**, **410** can be detached in any order or even simultaneously. In a first alternative, step **640** takes place chronologically before step **650**. In a second alternative, step **650** takes place chronologically before step **640**.

(40) Since the conduit **310** is preferably designed as a tube which extends in the interior **215** at least over 70% of the axial interior length **216** of the interior **215**, the first alternative is preferred for reasons of simpler handling.

(41) In a step **660**, sealing of the cartridge head duct **225** in terms of fluid conduction preferably takes place, in particular chronologically after the removal of the conduit **310** in step **650**. This allows mixing of the bone cement powder **500** and monomer liquid **510** under negative pressure, which can reduce air inclusions in the bone cement paste **520**.

(42) The cartridge head duct **225** is preferably sealed by means of a duct closure **270**.

(43) In a step **670**, a mixing of bone cement powder **500** and monomer liquid **510** preferably takes place in the interior space **215** of the cartridge **210** of the mixing unit **200**. For this purpose, it is preferred that when negative pressure is applied, the monomer liquid **510** is mixed with the bone cement powder **500** by means of the mixing rod **230** to form the bone cement paste **520**. The mixing rod **230** is preferably equipped with the mixing element **240**, preferably in the form of a mixing disk, in order to facilitate mixing. By mixing with applied negative pressure, air inclusions are reduced in the prepared bone cement paste **520**.

(44) In order to apply the prepared bone cement paste **520** from the device **100** to a desired location, it is preferred that, in an optional step **680**, the mixing rod **230** is pulled out such far out of the mixing rod duct **235** out of the interior space **215** until the mixing element **240** abuts distally on the dispensing plunger **280**, the part of the mixing rod **230** protruding proximally from the dispensing plunger **280** is subsequently broken off at a predetermined breaking point and the

dispensing plunger **280** is advanced out of the device **100** in the direction of the cartridge head closure **220** while the bone cement paste is dispensed. The advancement of the dispensing plunger **280** in the direction of the cartridge head closure **220** is preferably triggered with the aid of a dispensing aid **550**, such as a dispensing gun. This makes it easier for the user to use the device **100**.

REFERENCE SIGNS

(45) **100** Device **200** Mixing unit **210** Hollow-cylindrical cartridge **211** Proximal cartridge end **212** Distal cartridge end **213** Longitudinal axis of the cartridge **215** Interior space of the cartridge **216** Interior length **220** Cartridge head **225** Cartridge head duct **226** Cartridge head closure **230** Mixing rod **235** Mixing rod duct **240** Mixing element **250** Handle **260** Vacuum connection **270** Duct closure **280** Dispensing plunger **290** Conduit duct **300** Reservoir **310** Conduit **320** Reservoir container **325** Deformable region **330** Ampoule **331** Ampoule body **332** Ampoule head **333** Ampoule neck **340** Cavity **345** Filter element **350** Connection **355** Connection diameter **360** Connecting element **361** Clasp notches **362** Clasp element **400** First form closure **401** Straight line through first form closure **410** Second form closure **411** Straight line through second form closure **420** Pivot point **421** Straight line through pivot point **450** Tube **460** Negative-pressure source **500** Bone cement powder **510** Monomer liquid **520** Bone cement paste **530** Dispensing spout **550** Dispensing aid **600** Method **610** Opening in terms of fluid conduction **620** Flowing **630** Conveying **640** Detaching of second form closure **650** Detaching of first form closure **660** Closing in terms of fluid conduction **670** Mixing **680** Dispensing

Claims

1. A device for preparing a bone cement paste from a first starting component and a second starting components, comprising a mixing unit comprising a hollow-cylindrical cartridge with an interior space in which a bone cement powder is stored as the first starting component, a dispensing plunger which is displaceable axially in the interior space and seals the interior space at a proximal cartridge end in terms of fluid conduction, and a mixing rod which is guided into the interior space via a mixing rod duct of the dispensing plunger and is mounted movably in the interior space, and a reservoir for a monomer liquid as the second starting component, comprising a conduit that connects the interior space of the mixing unit to the reservoir in terms of fluid conduction, wherein the reservoir comprises a reservoir container, in which at least one ampoule, closed in terms of fluid conduction, having an ampoule body and an ampoule head, is arranged, and the monomer liquid is stored in the at least one ampoule, and a cavity in a region of the ampoule head, wherein the cavity is connected to the conduit in terms of fluid conduction and comprises a connection to the at least one ampoule, wherein the ampoule head is arranged at least in regions in the connection, and the reservoir container comprises a deformable region at least in portions so that tilting of the ampoule about a pivot point against the connection is enabled, wherein the conduit extends through a conduit duct of the dispensing plunger into the interior space.
2. The device in accordance with claim 1, wherein the conduit and the conduit duct form a first form closure between the mixing unit and the reservoir.
3. The device in accordance with claim 2, wherein the reservoir comprises a connecting element to connect the reservoir to the mixing unit via a second form closure.
4. The device in accordance with claim 3, wherein the connecting element is a clasp.
5. The device in accordance with claim 3, wherein the pivot point, the first form closure, and the second form closure form the vertices of a triangle in a side view.
6. The device in accordance with claim 3, wherein the first form closure, the second form closure, and the pivot point respectively lie on a straight line running parallel to a longitudinal axis of the cartridge, wherein the straight lines have a different straight-line distance from the longitudinal axis of the cartridge.

7. The device in accordance with claim 3, wherein the second form closure is formed between the connecting element and the mixing rod.
 8. The device in accordance with claim 1, wherein a vacuum connection is arranged on the dispensing plunger, via which vacuum connection the interior space is connected, in terms of fluid conduction, to a negative-pressure source.
 9. The device in accordance with claim 1, wherein the conduit is a tube and/or a hose.
 10. The device in accordance with claim 9, wherein the conduit is axially displaceable in the conduit duct.
 11. The device in accordance with claim 9, wherein the conduit extends at least over 70% of an axial interior length of the interior space.
 12. The device in accordance with claim 1, wherein the mixing unit comprises a duct closure in order to seal the conduit duct in terms of fluid conduction after removal of the conduit.
 13. The device in accordance with claim 1, wherein the cartridge comprises a cartridge head at a distal cartridge end axially opposite the proximal cartridge end, which cartridge head seals the distal cartridge end in terms of fluid conduction, wherein the cartridge head comprises a cartridge head duct for connecting, in terms of fluid conduction, a dispensing spout, via which the bone cement paste is dispensed from the interior space after the preparation and wherein the cartridge and the cartridge head are designed in one piece.
 14. A method for preparing a bone cement paste from two starting components by means of a device in accordance with one of the preceding claims, comprising the steps of: a. opening, in terms of fluid conduction, the at least one ampoule by tilting the ampoule, stored in the reservoir container, about the pivot point against the connection, b. flowing of the monomer liquid from the at least one ampoule into the cavity, c. conveying the monomer liquid from the cavity via the conduit into the interior space.
 15. The method in accordance with claim 14, additionally comprising the steps of: d. detaching the second form closure, e. detaching the first form closure by pulling the conduit out of the conduit duct, f. sealing, in terms of fluid conduction, the conduit duct, g. mixing the bone cement powder and the monomer liquid.
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